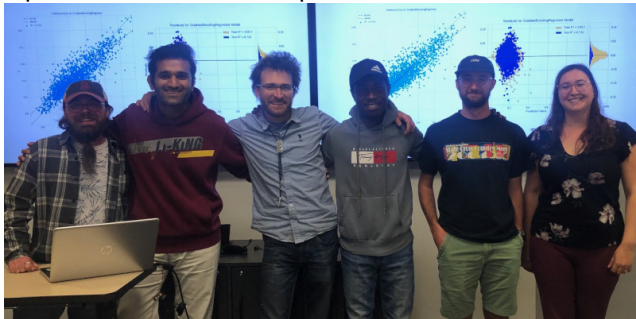


SOAK: Same/Other/All K-Fold Cross-Validation for Estimating Similarity of Patterns in Data Subsets

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Introduction: how to evaluate machine learning algorithms?

Proposed new algorithm, SOAK: Same/Other/All K-fold cross-validation

Results

Conclusions

Learning two different functions using two data sets

Figure from chapter by Hocking TD, *Introduction to machine learning and neural networks* for book *Land Carbon Cycle Modeling: Matrix Approach, Data Assimilation, and Ecological Forecasting* edited by Luo Y (Taylor and Francis, 2022).

Learning Algorithm	Train data	Learned function	Predictions on test data
--------------------	------------	------------------	--------------------------

Learn(		) → g	$g(\text{0}) = 0$
			$g(\text{1}) = 1$
			$g(\text{7}) = 1$

Learn(		) → h	$h(\text{shirt}) = 0$
			$h(\text{shirt}) = 0$
			$h(\text{pants}) = 1$

Learn is a learning algorithm, which outputs g and h .

Q: what happens if you do $g(\text{shoe})$, or $h(\text{circle})$?



Train on one subset and accurately predict on another?

- ▶ Predicting medical diagnosis (Lindly *et al.*), train on one year of surveys, test on another. (**different time periods**)
- ▶ Algorithmic fairness: predicting mortality (Laribi *et al.*), train on men, test on women. (**different sex, age, etc.**)
- ▶ Predicting trees/burn/fish/carbon emissions (Aslam, Shenkin, Thibault, Bodine, *et al.*), train on one geographic area, test on another. (**different geographic regions**)
- ▶ Cybersecurity: predicting vulnerabilities or attacks in different software/companies/etc. (**different technologies**)
- ▶ These are questions about **generalization**: how accurate is the learned function on a new/test data subset which is **qualitatively different** in some respect?
- ▶ “Should be accurate” if test data are similar to train data... how to check? Proposed SOAK algorithm.

Introduction: how to evaluate machine learning algorithms?

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Example data with subsets: predicting childhood autism

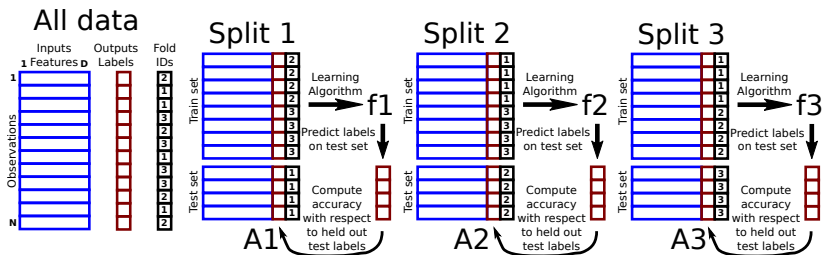
- ▶ Collaboration with Lindly *et al.*
- ▶ Downloaded National Survey of Children's Health (NSCH) data, years 2019 and 2020, from <http://www2.census.gov/programs-surveys/nsch>
- ▶ One row per person ($N = 46,010$ rows), one column per survey question ($D = 366$ columns).
- ▶ Output/label column is diagnosis with Autism (binary classification, yes or no),
can we predict it using the 365 inputs/features?
- ▶ 18,202 rows for 2019; 27,808 rows for 2020.

Proposed SOAK algorithm is a generalization of standard K-fold cross-validation, that can be used to answer two related questions:

- ▶ Can we train on one year, and accurately predict on another?
- ▶ Can we get a more accurate model by combining data from different years?

K-fold cross-validation: a standard algorithm used to estimate prediction accuracy in machine learning

- ▶ $K = 3$ folds shown below: three different models trained, three different prediction/test accuracy rates computed.
- ▶ Important: several train/test splits allow computation of P-values, to test for statistically significant differences between algorithms (not possible with only one split).



Hocking TD *Intro. to machine learning and neural networks* (2022).

Proposed SOAK algorithm (Autism data example)

- ▶ Example: childhood autism prediction data set.
- ▶ Subsets of interest are years, which can be represented by adding a new column to the data table.

	Inputs		Autism		Year
	Questions	D	Fold	Year	
People	1		1	2019	
			2	2019	
			3	2019	
			1	2020	
			2	2020	
			3	2020	

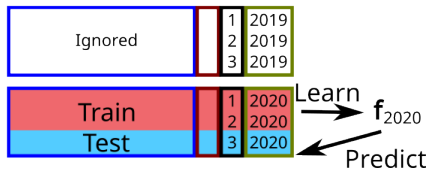
Proposed: Fold and Year are two different columns.
(In random Fold assignment, stratify by Year,
so there are equal numbers of rows of each fold in each year)

Proposed SOAK algorithm (Autism data example)

- ▶ Train subset same as test (=regular K -fold CV on 2020).
- ▶ Do we get a more or less accurate model than this baseline
Same model? (if we train on Other/All years)

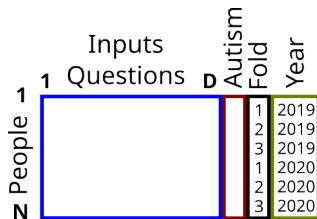
People	Inputs		Autism	Fold	Year
	1	Questions			
z	1			1	2019
				2	2019
				3	2019
				1	2020
				2	2020
				3	2020

Same

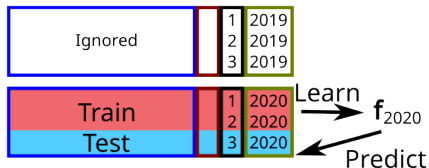


Proposed SOAK algorithm (Autism data example)

- ▶ Test subset fixed (2020), train on other subset/year (2019).
- ▶ Can we train on one year, and accurately predict on another? Compare Same/Other test error.



Same



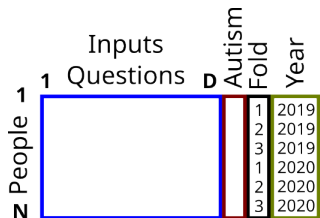
Other



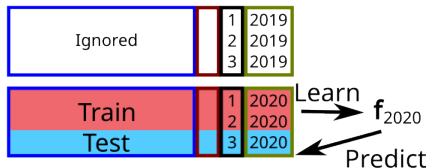
— Is Other as accurate as Same?

Proposed SOAK algorithm (Autism data example)

- ▶ Train set includes data from both subsets/years (2019, 2020).
- ▶ Can we get a more accurate model by combining data from different years? Compare Same/All test error.



Same



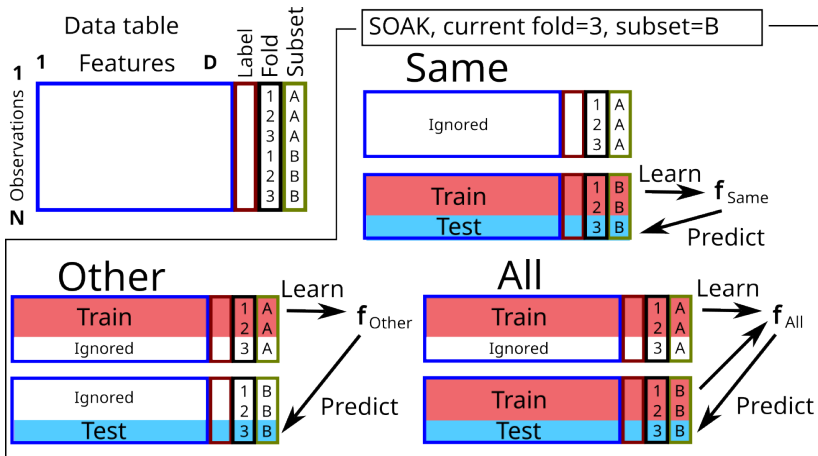
All

— Is All as accurate as Same?



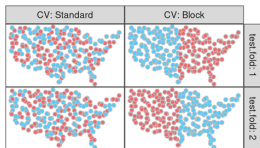
Proposed SOAK algorithm (generic data)

- ▶ Key new idea is **subset** column in data table.
- ▶ Example: $K = 3$ folds, two subsets (A/B).
- ▶ Compute test error for each fold (1/2/3) and subset (A/B).



Proposed SOAK algorithm (geographical data example)

Previous: standard and blocked train/splits



Significant difference between Standard and block CV?

— Is predicting more difficult when folds are assigned spatially rather than randomly?

- Un-paired P-value (less powerful).
- Block variable (West/East) used as fold.
- Number of folds must be equal to number of blocks (K=2 folds here for two subsets, West and East).

Valavi et al., 2019; Ploton et al., 2020; Schratz et al., 2024.

Proposed: SOAK train/test splits for test subset=West



Significant difference between Same and Other?

Significant difference between Same and All?

— How can we get best prediction in the West?

- Paired P-value (more powerful).
- Fold and subset are two different variables in data.
- Any number of folds possible (K=4 here for example).

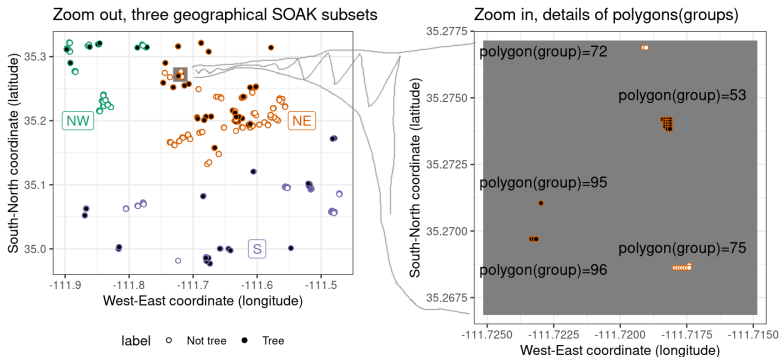
Previous (2 for loops) and proposed (3 for loops)

```
for fold_assignment_method in Random,Block: (previous)
  for test_fold from 1 to K:
    train_data = data[fold!=test_fold]
    test_data = data[fold==test_fold]
    compute mean test error over folds
compute P-value, Random-Block
```

```
for test_subset in A,B: (Proposed SOAK)
  for train_subsets in Same,Other,All:
    for test_fold from 1 to K:
      train_data = data[fold!=test_fold, train_subsets]
      test_data = data[fold==test_fold, test_subset]
      compute mean test error over folds
    compute P-values, Same-Other and Same-All
return P-values for each test subset
```

Proposed SOAK algorithm new to ML frameworks

- ▶ Previous scikit-learn/mlr3: when splitting, rows in a **group** stay together, **strata** proportions preserved.
- ▶ For example (below), satellite image segmentation, trees vs background (**strata**), Shenkin *et al.*: rows=pixels, **grouped by polygon**, SOAK **subsets** are geographical regions (NW, NE, S).
- ▶ SOAK: good predictions on one test **subset**, after training on Same/Other/All subsets? (can use together with groups/strata)



Previous work and novelty of SOAK

- ▶ scikit-learn (Pedregosa 2011) and mlr3 (Bischl 2019) support group/strata columns, **not subset**.
- ▶ Spatial/blocked cross-validation similar to SOAK “Other” model but **without distinct subset column, only one split per test block** (Valavi *et al.*, 2019; Ploton *et al.*, 2020; Schratz *et al.*, 2024).
- ▶ Same/Other comparison **without P-values** in out-of-distribution predictions in computer vision (Madan *et al.*, 2020).
- ▶ Farahani *et al.* (2021) review domain adaptation, Other model is “unsupervised” because no labels available in the target domain (**missing comparison with Same**).
- ▶ **Proposed SOAK (2026): new subset column distinct from previous fold/group/stratum columns, several train/test splits per test subset, P-values for Same/Other/All comparisons.**

Introduction: how to evaluate machine learning algorithms?

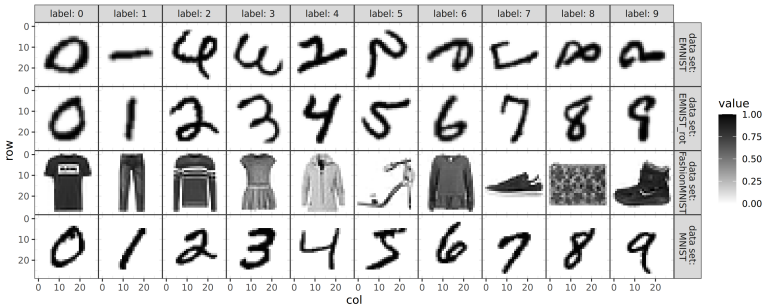
Proposed new algorithm, SOAK: Same/Other/All K-fold cross-validation

Results

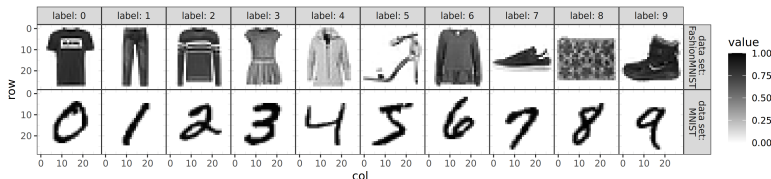
Conclusions

ImagePair data: train on MNIST and accurately predict on MNIST variants?

- ▶ Expect: $g(\text{img})$ and $h(\text{img})$ are not accurate.
- ▶ But train on MNIST, predict on EMNIST_rot (upright digits) should be possible, right? (Vote!) Let's use SOAK to find out!
- ▶ Create three new IPair data sets by combining MNIST with variants: EMNIST, EMNIST_rot, FashionMNIST.
- ▶ Each data set has two subsets: MNIST/variant.

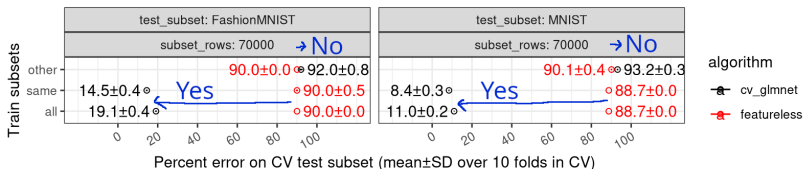


SOAK for IPair_Fashion data set (MNIST+FashionMNIST)



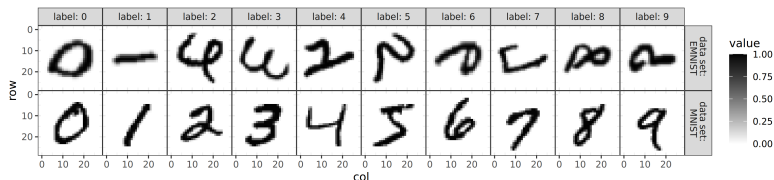
Data set: IPair_Fashion (MNIST+FashionMNIST)

Learning anything useful?



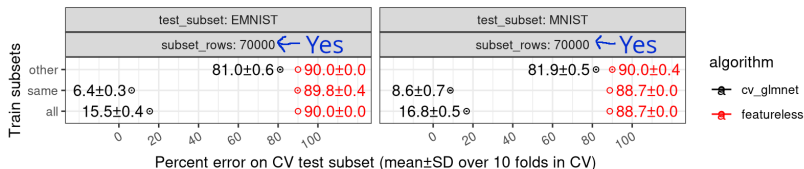
- ▶ **cv_glmnet** is L1-regularized linear model.
- ▶ **featureless** baseline always predicts most frequent class.
- ▶ **Other cv_glmnet** has greater test error than **featureless**.
- ▶ The patterns in the two subsets are too different for **Other cv_glmnet** to learn anything useful for prediction. (expected)

SOAK for IPair_E data set (MNIST+EMNIST)



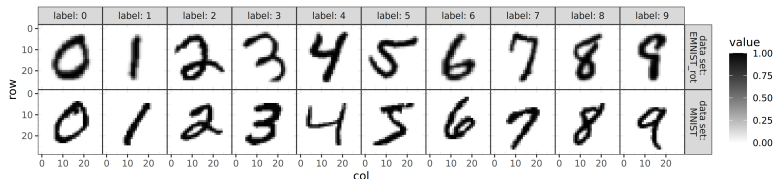
Data set: IPair_E (MNIST+EMNIST)

Learning anything useful?



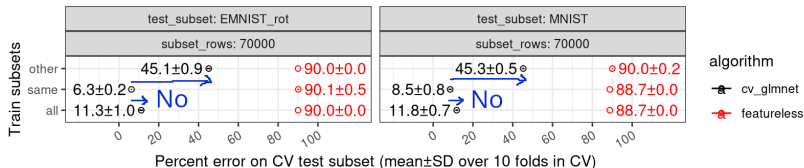
- ▶ **Other** **cv_glmnet** has significantly smaller test error than **featureless**, so linear model learns something that useful for predicting on the other subset.
- ▶ But since **Other** error rate is much larger than **Same**, clearly the pattern is very different between MNIST/EMNIST.

SOAK for IPair_E_rot data set (MNIST+EMNIST_rot)



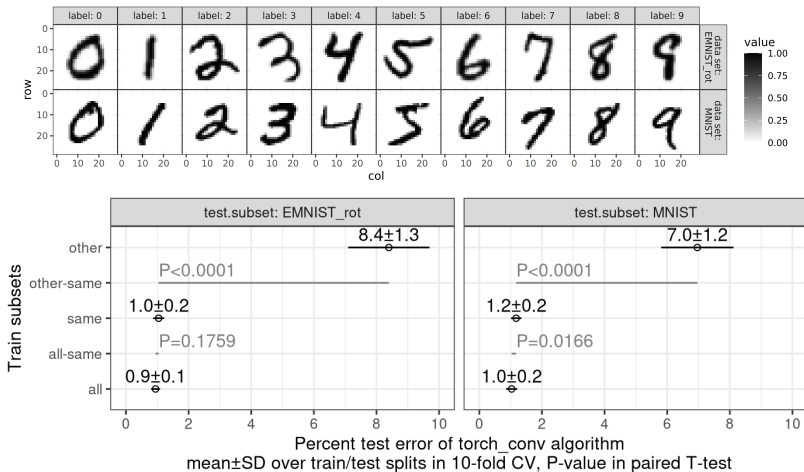
Data set: IPair_E_rot (MNIST+EMNIST_rot)

Other/All as accurate as Same?



- ▶ **Other** **cv_glmnet** has even smaller test error, indicating even more similarity between MNIST and EMNIST_rot data sets.
- ▶ But **Other/All** test error are still not as small as **Same**.
- ▶ Significant difference between patterns learnable/predictable by linear model in MNIST/EMNIST_rot. (surprising)

SOAK for IPair_E_rot data set (MNIST+EMNIST_rot)

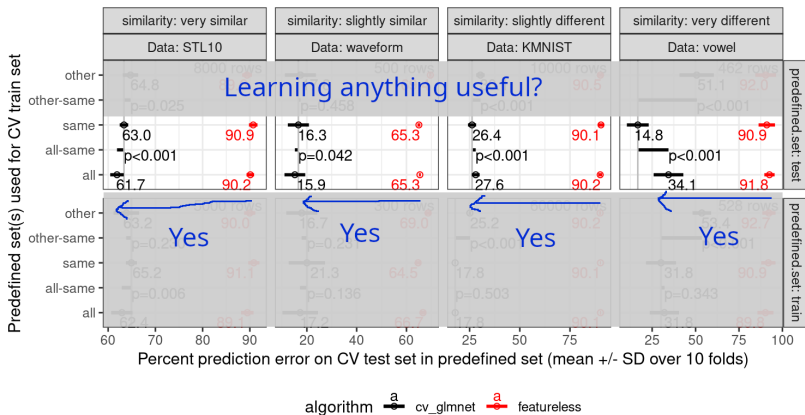


- ▶ For conv. net, test error for **Other** is still larger than **Same** (can't train on one subset and predict optimally on the other).
- ▶ Test error for **All** is smaller than **Same** — can combine subsets when training, for more accurate predictions.

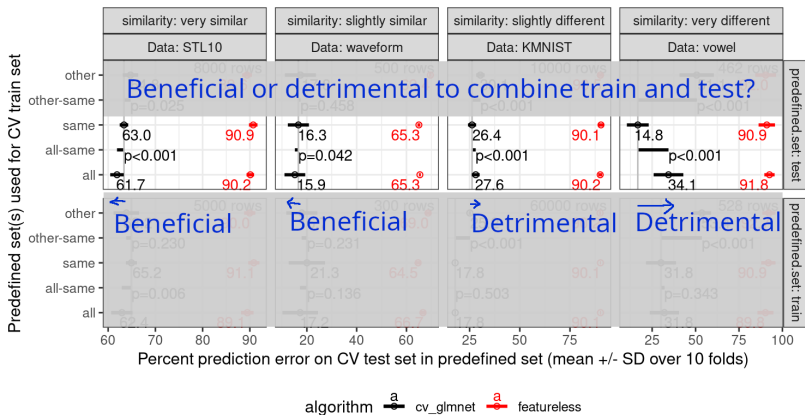
Benchmark data with pre-defined train/test subsets

- ▶ Machine learning researchers evaluate new algorithms using benchmark data sets, which sometimes have pre-defined train/test subsets.
- ▶ For example KMNIST is an image classification data set with 60,000 images in a pre-defined train subset, and 10,000 images in a pre-defined test subset.
- ▶ STL10 is another image classification data set with 5000 images in a pre-defined train subset, and 8000 images in a pre-defined test subset.
- ▶ Are the learnable/predictable patterns in the pre-defined train/test subsets similar? (expected if random sampling was used to create pre-defined subsets)
- ▶ Use pre-defined train/test subsets in SOAK, to see if patterns are learnable/predictable across pre-defined subsets.

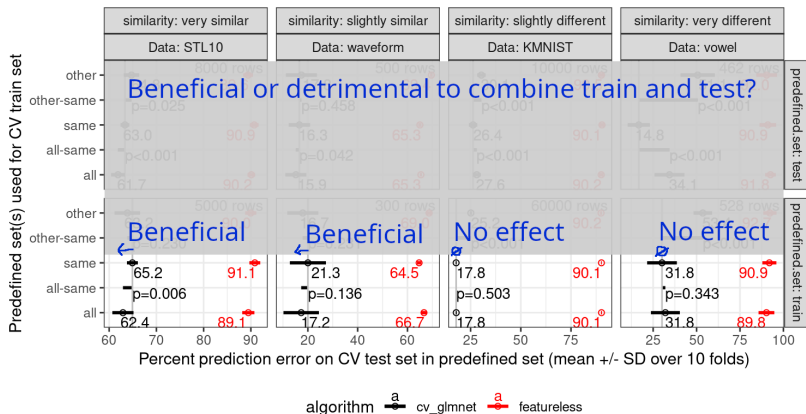
Benchmark data with pre-defined train/test subsets



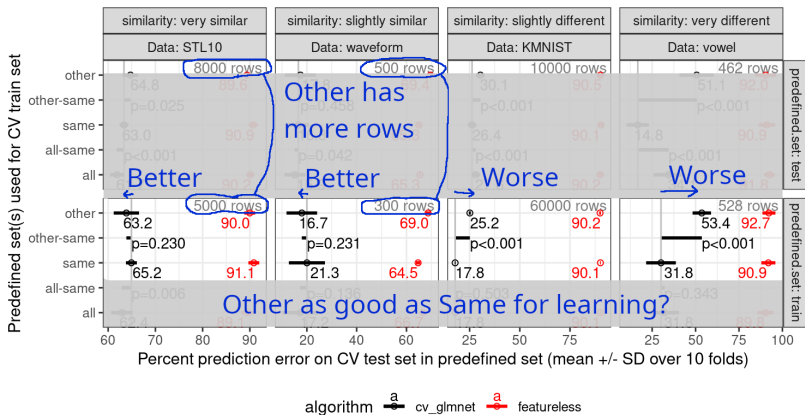
Benchmark data with pre-defined train/test subsets



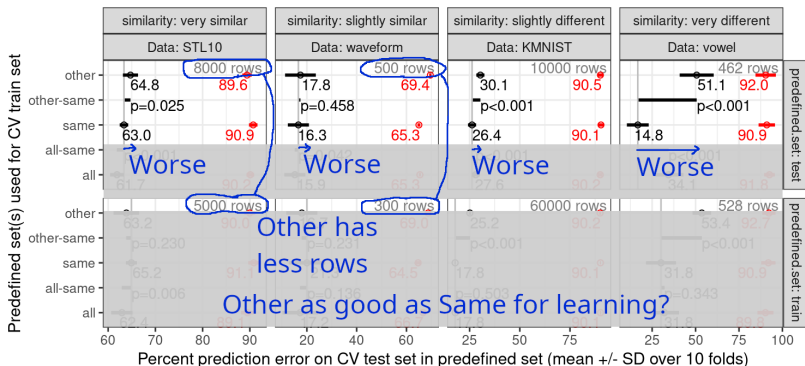
Benchmark data with pre-defined train/test subsets



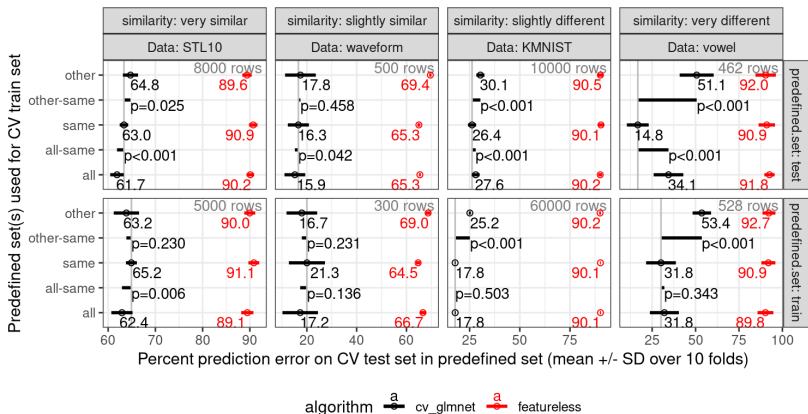
Benchmark data with pre-defined train/test subsets



Benchmark data with pre-defined train/test subsets



Benchmark data with pre-defined train/test subsets



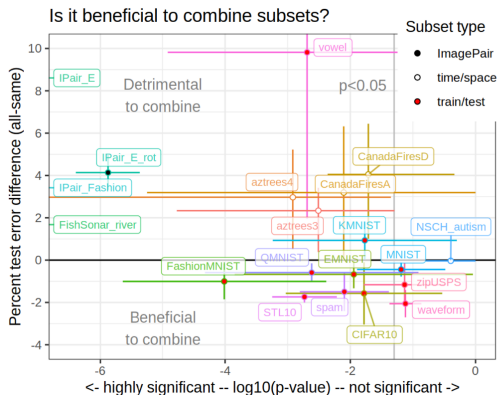
- ▶ Left data sets have similar pre-defined train/test subsets (expected), All error is always less than Same, and Other error is either greater or less than Same, depending on subset sizes.
- ▶ Right data sets have different pre-defined train/test subsets (surprising), All error is always greater than or equal to Same, and Other error is always greater than Same.

20 classification data sets analyzed using SOAK

We considered MNIST and variants as subsets (**ImagePair**), data from collaborations (**time/space** subsets), and benchmark data with pre-defined **train/test** subsets.

	Type	Data	rows	features	classes	subsets	imb.
1	● ImagePair	IPair_E	140000	784	10	2	1.0
2	● ImagePair	IPair_E_rot	140000	784	10	2	1.0
3	● ImagePair	IPair_Fashion	140000	784	10	2	1.0
4	○ time/space	CanadaFiresA	4827	46	2	4	7.0
5	○ time/space	CanadaFiresD	1491	46	2	4	1.6
6	○ time/space	FishSonar_river	2815744	81	2	4	1.2
7	○ time/space	NSCH_autism	46010	364	2	2	1.5
8	○ time/space	aztrees3	5956	21	2	3	2.0
9	○ time/space	aztrees4	5956	21	2	4	4.9
10	● train/test	CIFAR10	60000	3072	10	2	5.0
11	● train/test	EMNIST	70000	784	10	2	6.0
12	● train/test	FashionMNIST	70000	784	10	2	6.0
13	● train/test	KMNIST	70000	784	10	2	6.0
14	● train/test	MNIST	70000	784	10	2	6.0
15	● train/test	QMNIST	120000	784	10	2	1.0
16	● train/test	STL10	13000	27648	10	2	1.6
17	● train/test	spam	4601	57	2	2	2.0
18	● train/test	vowel	990	10	11	2	1.1
19	● train/test	waveform	800	21	3	2	1.7
20	● train/test	zipUSPS	9298	256	10	2	3.6

All-Same comparison across 20 data sets



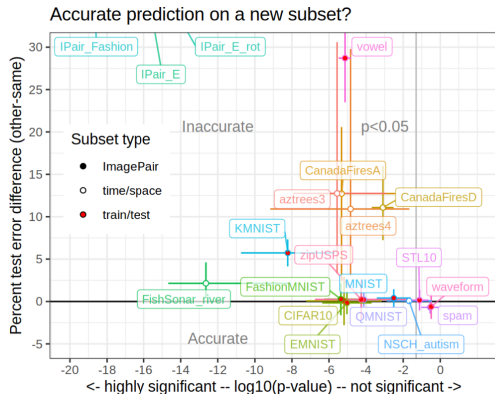
Data	ErrorDiff	log10(P)
● IPair_E	8.2, 9.1	-11.2, -10.2
● IPair_E_rot	3.3, 5.0	-6.4, -5.4
● IPair_Fashion	2.6, 4.6	-12.4, -8.6
● vowel	2.0, 17.7	-4.9, -0.5
○ CanadaFiresD	1.0, 6.4	-2.4, -0.3
○ aztrees4	0.5, 5.2	-7.0, -1.4
○ FishSonar_river	0.5, 4.4	-14.4, -8.6
○ aztrees3	0.4, 3.4	-4.8, -1.3
● KMNIST	0.0, 1.8	-3.2, -0.3
○ CanadaFiresA	-0.0, 6.3	-5.2, 0.0
○ NSCH_autism	-0.1, -0.0	-0.8, -0.0
● MNIST	-0.8, -0.1	-1.9, -0.5
● QMNIST	-1.0, -0.2	-4.3, -0.9
● EMNIST	-1.3, -0.0	-3.8, -0.0
● FashionMNIST	-1.9, -0.2	-5.6, -2.4
● STL10	-2.0, -1.5	-3.2, -2.2
● zipUSPS	-2.2, -0.1	-1.8, -0.5
● spam	-2.4, -0.6	-2.8, -1.4
● waveform	-2.7, -1.4	-1.4, -0.9
● CIFAR10	-3.0, -0.1	-3.0, -0.5

Figure: For each data set, point/segments show mean/range over test subsets.

Table: ranges over test subsets.

- ▶ Top: 10 data sets with incompatible subsets.
- ▶ Bottom: 10 data sets with compatible subsets.

Other-Same comparison across 20 data sets



Data	ErrorDiff	log10(P)
● IPair_Fashion	77.5, 84.7	-21.7, -17.8
● IPair_E	73.2, 74.7	-19.2, -18.3
● IPair_E_rot	36.7, 38.8	-15.3, -15.1
● vowel	23.5, 33.9	-5.5, -4.8
○ CanadaFiresD	7.2, 16.0	-3.7, -2.5
● KMNIST	4.1, 7.3	-10.7, -5.7
○ aztrees3	3.3, 30.6	-7.9, -2.9
○ aztrees4	3.0, 29.8	-9.2, -1.7
○ CanadaFiresA	2.4, 20.6	-6.7, -2.4
○ FishSonar_river	0.5, 4.6	-14.7, -8.1
○ NSCH_autism	-0.1, 0.2	-2.9, -0.5
● QMNIST	-0.6, 1.0	-4.9, -3.3
● MNIST	-0.6, 1.5	-3.4, -1.6
● STL10	-1.1, 1.4	-1.6, -0.6
● spam	-1.4, 0.1	-1.0, -0.0
● EMNIST	-1.5, 1.2	-6.4, -3.7
● FashionMNIST	-1.6, 2.2	-6.3, -4.5
● waveform	-2.0, 0.8	-0.6, -0.3
● zipUSPS	-2.1, 2.6	-6.8, -1.8
● CIFAR10	-2.8, 2.9	-7.2, -3.1

Figure: For each data set, point/segments show mean/range over test subsets.

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Introduction: how to evaluate machine learning algorithms?

Proposed new algorithm, SOAK: Same/Other/All K-fold cross-validation

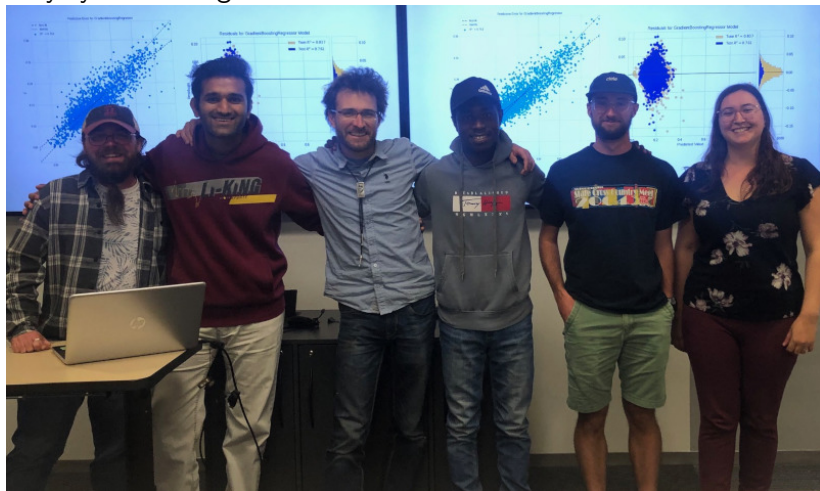
Results

Conclusions

Summary of contributions

- ▶ Proposed SOAK algorithm estimates similarity of learnable patterns between subsets (space, time, etc).
- ▶ SOAK idea is new to ML frameworks (proposed **subset** column/idea not the same as **group/stratum**).
- ▶ Free/open-source R package available in mlr3 framework (easy parallelization over algorithms, data sets, train/test splits) <https://github.com/tdhock/mlr3resampling>
- ▶ Ran SOAK in parallel over 20 classification data sets, 10 folds, 2–4 subsets, Same/Other/All (1200+ train/test splits).
- ▶ Train on MNIST, predict on FashionMNIST (impossible), predict on EMNIST (surprisingly difficult, both are digits).
- ▶ Most pre-defined train/test subsets in benchmark data are similar (STL10/waveform), some are not (KMNIST/vowel).
- ▶ We observed similarity between years in Autism data (slight benefit to combining years, predicting on new year works).
- ▶ In fires/trees/fish data, we observed significant differences between images/regions/rivers.

Thanks! Please email me if you are interested to collaborate:
toby.dylan.hocking@usherbrooke.ca



Reproducible slide/figure source code:

<https://github.com/tdhock/cv-same-other-paper>

<https://github.com/tdhock/max-generalized-auc>

Learning algorithms we consider

We used the following learning algorithms:

`cv_glmnet` L1-regularized linear model (feature selection).
Friedman, *et al.* (2010).

`xgboost` Extreme gradient boosting (non-linear). Chen and
Guestrin (2016).

`rpart` Recursive partitioning, decision tree (non-linear,
feature selection). Therneau and Atkinson (2023).

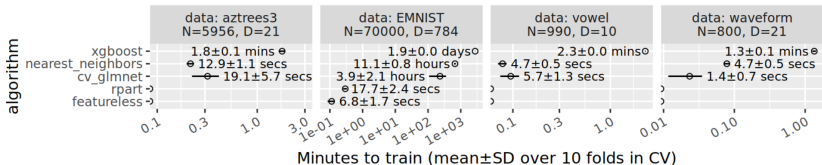
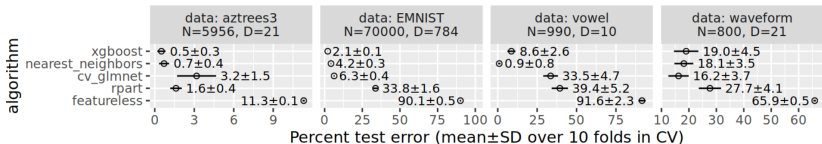
`nearest_neighbors` classic non-linear algorithm, as implemented in
knnn R package. Schliep and Hechenbichler (2016).

`featureless` un-informed baseline, ignores all inputs/features, and
always predicts the most frequent label in train data.
For example, Autism=No. Nomenclature from mlr3
R package, Lang, *et al.*, (2019).

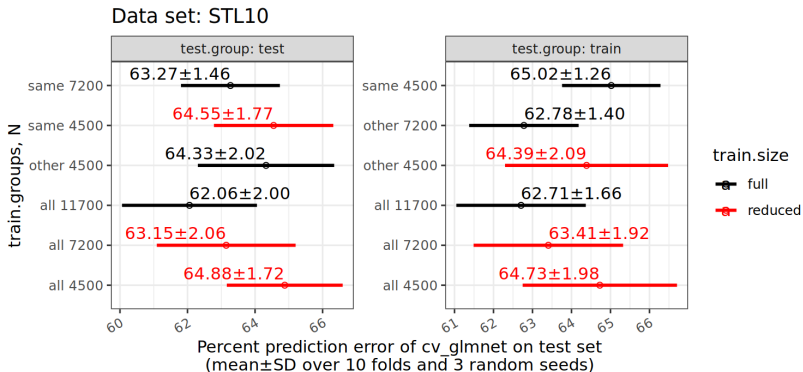
Each learning algorithm has different properties (non-linear, feature
selection, etc). For details see Hastie, *et al.* (2009) textbook.

Learning algorithms have different accuracy and speed

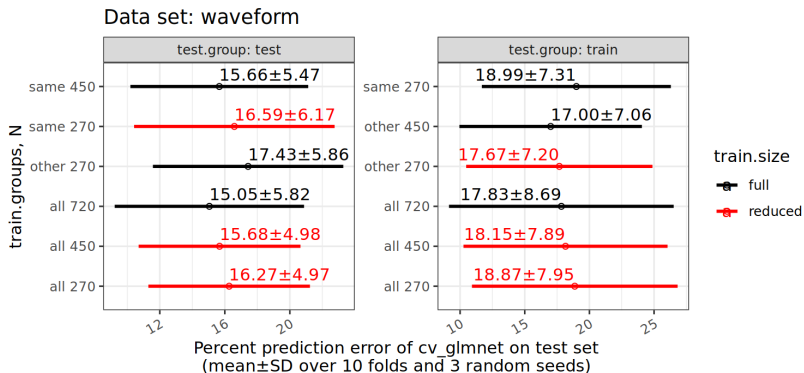
- ▶ Image segmentation/classification, and audio classification data sets.
- ▶ xgboost is slowest to train on all four data sets, most accurate in aztrees3, EMNIST.
- ▶ Other algorithms are faster and more accurate in vowel, waveform.
- ▶ Need to try a variety of algorithms for optimal prediction in any data.



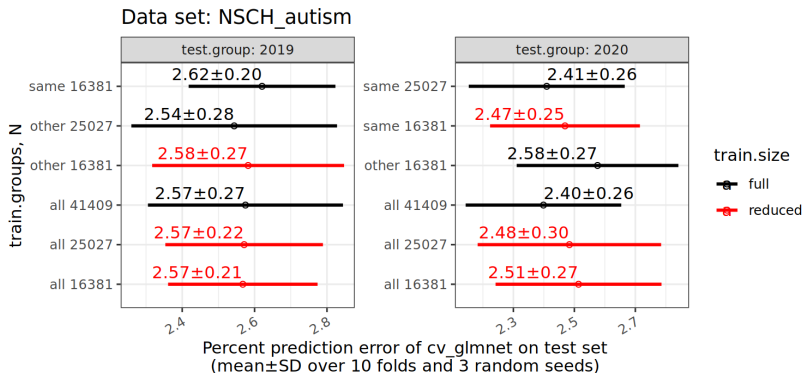
Large sample size effect



Moderate sample size effect



Small sample size effect



Stratified fold assignment, 10-fold CV

```
> folds <- function(by){set.seed(1);  
+ data.table(label=rep(0:1,c(100,900))) [  
+ , fold := sample(rep(1:10,l=.N)), by  
+ ][, table(label, fold)]}  
> folds()  
      fold  
label 1  2  3  4  5  6  7  8  9 10  
      0 13 13  4 10 13  7 12  8  9 11  
      1 87 87 96 90 87 93 88 92 91 89  
> folds("label")  
      fold  
label 1  2  3  4  5  6  7  8  9 10  
      0 10 10 10 10 10 10 10 10 10 10  
      1 90 90 90 90 90 90 90 90 90 90
```

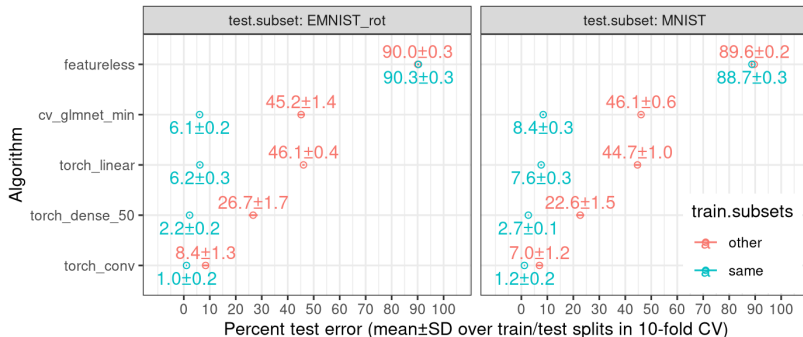
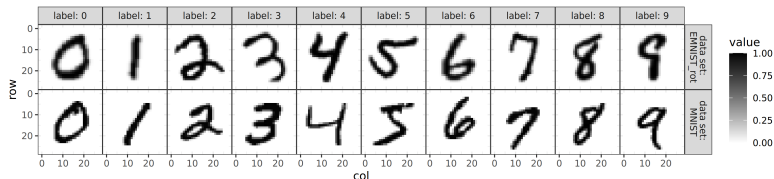
With stratification by label, we get the same label counts in each fold.

ML framework feature comparison

Language Framework	R mlr3	how?	python scikit-learn
Learners	multiple	benchmark_grid	single
Parallelization	yes	mlr3batchmark	no
Subsets(proposed)	yes	mlr3resampling	no

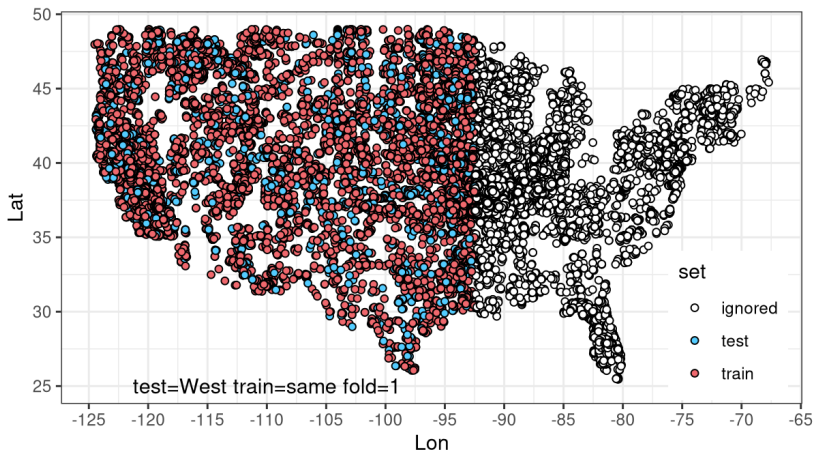
- ▶ mlr3 supports parallelization over algorithm, data sets, and train/test splits via mlr3batchmark package (ex: SLURM).
- ▶ scikit-learn cross_validate computes test error for a single algorithm (no special parallelization support).
- ▶ For an example of how to use stratum, subset, and group in mlr3, see “Arizona trees data” section of https://cloud.r-project.org/web/packages/mlr3resampling/vignettes/Newer_resamplers.html
- ▶ Scikit-learn https://scikit-learn.org/stable/modules/cross_validation.html#stratifiedgroupkfold supports group, and stratification by class, but not subset.

SOAK for IPair_E_rot data set (MNIST+EMNIST_rot)



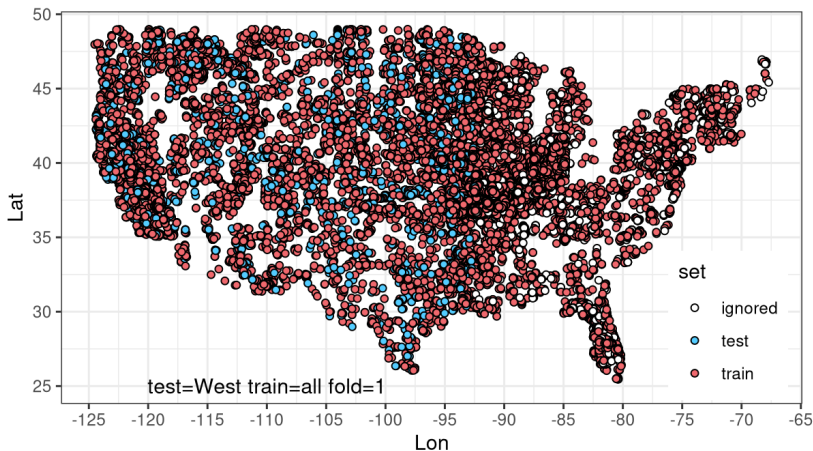
Convolutional neural network (torch_conv) has smaller test error than linear model; **Same** still has smaller test error than **Other**.

Proposed SOAK algorithm (geographical data example)



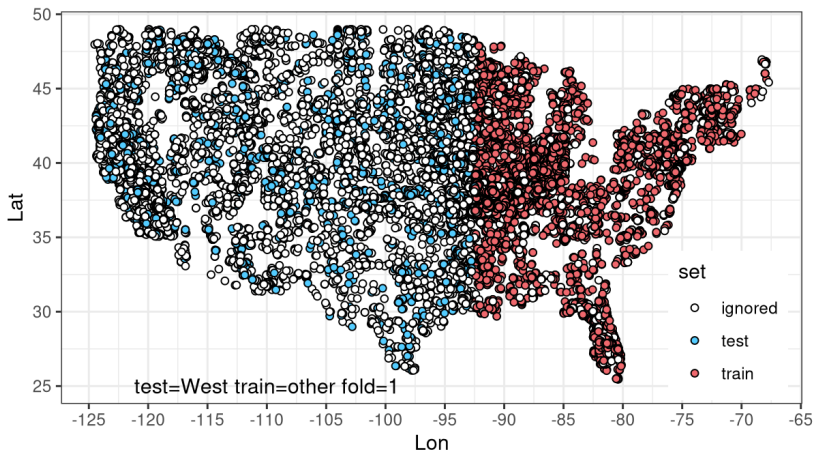
- ▶ Say we want good prediction in western USA.
- ▶ One of the K test sets is: subset=West and fold=1.
- ▶ Train on Same (West, fold $\neq$ 1) establishes a baseline.

Proposed SOAK algorithm (geographical data example)



- ▶ Train on All (West+East, $\text{fold} \neq 1$) would be less accurate than Same if subsets are different, and more accurate if subsets are similar (more relevant training data).

Proposed SOAK algorithm (geographical data example)



- ▶ Train on Other (East, $\text{fold} \neq 1$) would be less accurate than Same if subsets are different.
- ▶ If subsets are similar, then the larger of Same and Other should be more accurate (more relevant training data).